Report on - Hands-on Session on Building a 4-Wheeled Robot

Title: Hands-on Session on Building a 4-Wheeled Robot
Date: 14th February 2026
Time: 11am - 1pm
Venue: RF Lab , B-Wing ,First Floor

Target Audience: SE EXTC

No. of Participants Present: 5

No. of Girl Participants Present: 1

No. of Boy Participants Present: 4

Organizing Department / Committee / Authority: IEEE - RAS

Faculty Coordinator: Mrs. Freda Carvalho

Objectives:
- - Introduce participants to the fundamentals of Arduino programming and hardware setup.
- - Demonstrate Bluetooth communication using the HC-05 module for wireless control.
- - Guide attendees through the design and construction of a four-wheel robotic bot.

Outcomes:
- Participants developed confidence in working with microcontrollers and wireless modules.
- The workshop fostered teamwork and problem-solving skills through collaborative bot-building.
- Attendees left with a practical foundation to pursue advanced robotics and IoT projects.

Detailed Report: The IEEE student branch organized a hands-on workshop **exclusively for IEEE-DBIT members** to introduce the basics of Arduino programming, HC-05 Bluetooth interfacing, and the construction of a four-wheel robotic bot. The session was designed to provide participants with practical exposure to embedded systems and robotics, encouraging them to apply theoretical knowledge in a collaborative environment. A total of five Second-Year Electronics and Telecommunication (SE EXTC) members attended the workshop, engaging actively with the demonstrations and exercises.

The workshop was led by Aryan Puranik and Daniel Sebastian, who guided participants through the technical aspects of coding, module integration, and bot assembly. Coordination and participant support were managed by Neha Haldankar and Samiksha Gaonkar, ensuring smooth execution and assisting attendees during practical activities. The event successfully enhanced participant's confidence in working with microcontrollers and wireless modules, fostered teamwork, and laid a strong foundation for future robotics and IoT projects within the IEEE community.

Snapshot of the Event:







Details of the Attendees:

S. No.	Name
1	Amey Kanade
2	Tanvi Khamitkar
3	Tirth Shetty
4	Kunal Jadhav
5	Aayush Borle

Report Prepared By: IEEE-DBIT

Name of the Student: Aarya Jadhav & Shreya Manoj

Post of the student: Reporting Head & Joint Reporting Head, IEEE-DBIT

Report Approved By: IEEE - DBIT

Name of the Faculty: Ms. Freda Carvalho

Post of the Faculty: IEEE-DBIT Student Branch Counsellor